

AI Mathematical Olympiad Competition

1 Introduction

In this project, students will engage in the task of building and improving the performance of Large Language Models (LLMs). The project is not restricted to previously learned techniques, but instead encourages students to flexibly apply their knowledge of data science and machine learning to analyze, evaluate, and enhance the mathematical problem-solving capabilities of these models.

Specifically, students will participate in the *AI Mathematical Olympiad – Progress Prize 3* competition (Kaggle competition link) hosted on the Kaggle platform. The competition consists of 110 mathematical problems spanning multiple domains, including algebra, combinatorics, geometry, and number theory, with difficulty levels ranging from national olympiad standards to International Mathematical Olympiad (IMO) level at the high school stage.

The objective of this project is to improve the performance of LLM-based systems in solving advanced mathematical problems. The dataset consists of problem statements written entirely in “Latex”, which requires students to handle unstructured data, perform feature analysis, and design appropriate modeling strategies.

2 Competition Data and Evaluation

2.1 Competition Dataset

The competition dataset consists of 110 problems, where the answer to each problem is a non-negative integer between 0 and 99999, inclusive. All problems are provided purely as text with mathematical notation written in “Latex” (students are encouraged to consult the **Data** section of the competition for detailed descriptions).

The competition provides a **Public Test Set** containing 50 problems for ongoing evaluation during the competition, and a **Private Test Set** containing another 50 problems for final evaluation. The remaining 10 problems are provided as reference materials in the accompanying files.

Note: Since this is a competition-based project, students must strictly follow the required submission format, as well as the prescribed usage of APIs and Python code structure specified in the official demo files provided by the competition organizers.

2.2 Competition Evaluation

Throughout the submission period, students' models will be evaluated on the **Public Test Set**, and performance will be measured by the number of correctly answered problems (out of 50). After the competition deadline, final evaluation will be conducted on the **Private Test Set** by the organizers, with minor adjustments to the scoring procedure (see the **Evaluation** section of the competition for details).

In addition, the competition enforces a submission limit of one submission per day (see the **Rules** section for more information). Students are therefore advised to plan their experiments and submissions carefully throughout the project duration to avoid last-minute submissions.

3 Project Requirements

3.1 Execution Environment

Students are required to conduct the project in teams registered in the corresponding theoretical course (with a maximum of 4 members per team). Each team must create a Kaggle team with a name (TeamName) that follows the required naming convention, using "HCMUS" as the prefix.

All experiments and submissions must be implemented in the form of Jupyter Notebooks, following the standard Kaggle notebook format and complying with the regulations specified by the competition organizers.

3.2 Requirements

In this project, students are allowed to use GPUs provided by the competition organizers to load open-source models and perform inference directly. The primary objective of the project is to optimize model performance in order to achieve the best possible results in the competition, which will be used for ranking and objective performance evaluation.

Teams are free to design their own pipelines and processing strategies according to their ideas, as long as the proposed approaches are reasonable and can lead to improved performance in the evaluation stages. Students are encouraged to experiment with different techniques, combine multiple methods, and select approaches that align with their knowledge background and research interests.

Some basic suggested directions include analyzing the characteristics of individual problems, encouraging models to explain incorrect predictions, and enabling models to learn from previous mistakes, among others.

Note: Since open-source models come in various sizes, students are encouraged to refer to publicly available code and notebooks from previous participants. The selected model configurations must fit within the maximum computational resources provided by the organizers, which is approximately ~ 80 GB of GPU memory. Example models include GPT-based models, Qwen 2.5, Qwen 3, DeepSeek-Math, and related variants.

3.3 Project Timeline

The project will be carried out throughout the semester, with two major milestones: the mid-term oral examination and the final oral examination. Before the final oral examination, each team must select a final version of their model for presentation and discussion.

In addition, teams are encouraged to conduct experiments and evaluations on a regular (daily) basis to assess model effectiveness, taking into account the competition's limitation on the number of submissions allowed per day.

4 Submission

Upon submission, student teams are required to submit all necessary materials, including the written report, source code (notebooks), and presentation slides, packaged into a single compressed ZIP file (with the `.zip` extension). The ZIP file must be named according to the format “Team-Name.zip”. If the submission file is too large, teams may upload the materials to Google Drive and include the download link in a text file, ensuring that the link is accessible and downloadable.

4.1 Source Code (Jupyter Notebook)

The source code and experimental results must be reported in the form of Jupyter Notebooks and must satisfy the following requirements:

- Include student information (student ID, full name, etc.).
- Provide clear explanations for each major component and detailed descriptions of the implementation steps, optionally supported by illustrative figures and diagrams..
- Ensure that all code sections are properly commented and that the overall notebook structure is logically organized.
- Ensure that the notebook has been fully executed (“Run All”) before submission.

4.2 Report

The detailed written report must include the following contents:

- Student information (student ID, full name, etc.).
- A self-assessment of each team member's contribution to the project.
- A clearly structured presentation in PDF format.
- A comprehensive description of the proposed approach, from the initial idea to the detailed implementation steps, together with the necessary background knowledge.
- An analysis of the strengths, weaknesses, and limitations of the proposed method.
- A complete list of references, formatted consistently.

4.3 Presentation Slides

Student teams are required to submit presentation slides for both the mid-term and final oral examinations. The slides must clearly present the project content and include essential information such as the team name and student details.

5 Assessment

No.	Component	Weight
1	Mid-term oral examination	10%
2	Final oral examination	40%
3	Report	30%
4	Competition ranking	20%
5	Bonus	up to 10%
	Total	100% + Bonus (max 10%)

Additional Notes:

- The competition ranking component is evaluated based on the relative ranking of student teams on the competition leaderboard, and teams are compared against one another solely based on their final results.

- Bonus points may be awarded based on additional criteria such as achieving strong performance with relatively small model sizes, originality of the proposed approach, or other notable contributions.

6 Notices

- This is a group-based assignment, and all student teams must comply with the submission rules specified in this document.
- The project is conducted throughout the entire semester.
- Any form of academic dishonesty, including cheating, plagiarism, or late submission, will result in a score of zero for the entire course.

Good luck.